

Communication-Matched Warp Collectives for ML-KEM and ML-DSA on a RISC-V SIMT GPU

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Abstract—ML-KEM and ML-DSA share Keccak and number-theoretic transforms (NTTs), yet their useful parallelism has incompatible communication: Keccak repeatedly reduces and permutes a 25-word state, while an NTT exchanges coefficient pairs at power-of-two distances. We present communication-matched warp collectives that keep both states in ordinary lane registers of an RV32 RISC-V GPU. The instructions are stateless, use standard source vectors and single-destination writeback, and implement fixed Keccak neighborhoods or paired NTT butterflies. A serialized modular unit shares its routing and multiplier bank across the 16-bit ML-KEM and 32-bit ML-DSA datapaths. A 200-MHz V80 evaluation covers all six standardized parameter sets against a matched, optimized cooperative software mapping. Stage, shared NTT, and arithmetic reuse accelerate complete single-request ML-KEM by $2.314\text{--}2.365\times$ and ML-DSA by $1.910\text{--}2.150\times$; eight-request batches gain $1.914\text{--}2.126\times$. A separate five-point bank sweep selects two multipliers: it removes 87.500% of NTT-hierarchy DSPs at a 2.963% NTTBF test-cycle and at most 0.143% request penalty in middle-set Pointer controls. The results show that GPU cryptographic hardware must be sized at request scope, not by primitive peak throughput.

Index Terms—post-quantum cryptography, ML-KEM, ML-DSA, Keccak, NTT, SIMT, RISC-V

I. INTRODUCTION

The standardization of ML-KEM and ML-DSA gives post-quantum key establishment and digital signatures a common deployment target [1], [2]. Although the two schemes provide different services, both repeatedly invoke SHA-3/SHAKE and polynomial arithmetic. Efficient implementations must also execute sampling, encoding, and, for signatures, input-dependent rejection loops. Accelerating an isolated transform or permutation is therefore insufficient: the useful outcome is lower latency or higher throughput for a complete cryptographic request, including the surrounding computation.

A programmable GPU offers two levels of parallelism for these workloads. Threads can cooperate within a polynomial or a hash state, while independent warps process different requests. GPU implementations of Kyber and Dilithium already exploit this structure through register reuse, kernel fusion, batching, and cooperative scheduling [3], [4]. ML-Cube further combines GPU execution and machine-learning accelerators, including a 25-thread Keccak mapping [5]. These results motivate a complementary question for an extensible SIMT processor: which remaining work merits specialized hardware once the software already exploits warp parallelism?

Our starting point is the cost shared by both schemes. In a historical single-lane RV32 software study, removing the Keccak permutation bodies reduces a complete ML-KEM-768

request by 65.65%, while independently removing NTT/INTT bodies reduces it by 12.67%. ML-DSA key generation shows the same ordering (Section II). These diagnostics identify two reusable targets; they do not predict the benefit over optimized GPU software. We therefore build a stronger comparison point: register-resident, warp-cooperative Keccak and NTT kernels, with parallel sampling, encoding, and auxiliary arithmetic enabled identically in every hardware comparison. Our reported gains measure specialization beyond that software mapping.

The obstacle is not a lack of arithmetic parallelism but the cost of making operands meet. Each NTT layer contains 128 independent butterflies. With coefficients distributed across lane registers, some partners lie in the same lane and others in lanes separated by an XOR distance. Keccak-f[1600] instead couples 25 64-bit words through column reductions, fixed permutations, and row-neighbor logic [6]. Generic shuffles express both patterns, but introduce instructions and dependencies at each exchange. Keeping the state in registers removes intermediate memory traffic without removing these communication costs. The two primitives thus favor different internal connections even when they share the same architectural register interface.

Prior hardware addresses parts of this problem at other boundaries. RISC-V software optimization [7] and finite-field instruction extensions [8], [9] reduce arithmetic cost; RISQ-V and Nannipieri et al. add coarser polynomial operations to scalar cores [10], [11]. Configurable lattice processors such as Sapphire and KaLi already reuse resources across protocols [12], [13], while KiD explicitly unifies Kyber/Dilithium NTT multiplication [14]. Full-protocol FPGA accelerators also exploit request batching [15]. Sharing arithmetic is therefore established; the issue here is how shared arithmetic consumes distributed SIMT operands and returns results through an existing GPU pipeline.

The hash interface presents a related choice. Prior work quantifies how state transfers and sponge input/output affect hardware hashing performance [16]. Recent RISC-V extensions retain hash state in private registers [17]; a published patent also describes a Keccak round executed collectively by a warp [18]. Building on these interfaces, we study the constraints of an RV32 SIMT core: handling 64-bit state with 32-bit destinations, preserving per-warp dependency tracking, and selecting the instruction granularity at complete-request scope. A faster permutation can have little effect once sampling, sponge I/O, or polynomial work limits execution.

We address three coupled design constraints. First, collective operations must fit ordinary issue and writeback: an

NTT butterfly logically has three inputs and two outputs, while a Keccak step mixes values from many lanes. Second, ML-KEM’s 16-bit and ML-DSA’s 32-bit coefficients require different reductions without duplicating the entire modular datapath. Third, provisioning all 16 pair products in parallel may waste resources when other instructions or warps hide their latency. Neither an isolated primitive benchmark nor a comparison against scalar C resolves these constraints.

We implement *communication-matched warp collectives* in Vortex [19]. The state remains in ordinary GPRs, and each instruction replaces a fixed communication step together with its arithmetic (Fig. 1). Two lanes jointly execute one NTT butterfly, compute one Montgomery product, and each receive one result. A shared serialized multiplier bank handles both moduli and also serves lane-local products. Keccak exposes either three stage operations or one fused round, with explicit low/high outputs on RV32. Every instruction returns a single architectural destination vector and retains no cryptographic state between instructions. We compare these register interfaces with a memory-coupled permutation engine and size the hardware using complete requests.

The resulting Stage-plus-NTT design, including modular-arithmetic reuse, accelerates complete single-request ML-KEM by $2.314\text{--}2.365\times$ and ML-DSA by $1.910\text{--}2.150\times$ over matched cooperative software on a 200-MHz Alveo V80. All six standardized parameter sets are covered. NTT specialization provides additional request-level gains after Keccak acceleration. A separate bank sweep finds that two multipliers replace 16 with at most 0.143% request-cycle increase in its measured workloads. Our contributions are:

- A stateless SIMT instruction interface for Keccak neighborhoods and XOR-paired NTT butterflies, with ordinary operand tracking and one destination per lane; controlled Stage, Round, and Pointer comparisons establish the importance of the execution boundary.
- A shared K/D modular unit that combines pair routing, modulus-specific reduction, and a serialized multiplier bank. It supports forward/inverse NTT and reuses the same arithmetic outside transforms without private polynomial storage.
- A complete-library, six-parameter FPGA evaluation with matched software baselines and separate Keccak, NTT, and arithmetic ablations. Concurrency and five-point multiplier sweeps connect primitive performance to request latency, batch throughput, and routed hardware cost.

II. BACKGROUND AND MOTIVATION

A. Shared hotspots and the scope of profiling

ML-KEM uses degree-256 polynomials modulo $q_K = 3329$; ML-DSA uses $q_D = 8380417$. Both use SHA-3/SHAKE for hashing and pseudorandom expansion, including polynomial sampling [1], [2], [6]. Table I reports our initial single-lane RV32 SimX diagnosis: one core, four warps of four threads, and no L2/L3. Each independent ablation removes one primitive’s body while preserving its call interface; the

TABLE I
INITIAL SOFTWARE COST DIAGNOSIS. INDEPENDENT ABLATION DELTAS, NOT ADDITIVE PHASE FRACTIONS OF THE CURRENT BOARD BASELINE.

Workload	Keccak	NTT/INTT
ML-KEM-768 complete request	65.65%	12.67%
ML-DSA-65 key generation (full RAM)	74.44%	7.62%

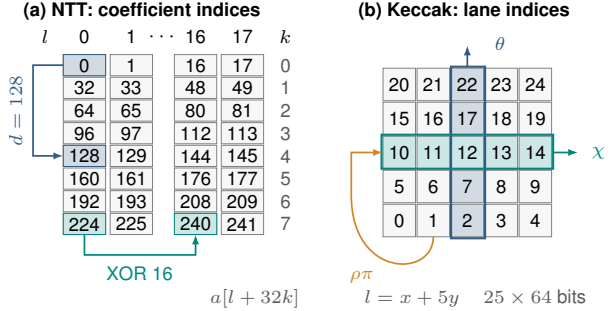


Fig. 1. Register placement determines the communication. (a) Selected NTT lanes: indices 0 and 128 pair within lane 0; 224 and 240 pair across lanes 0 and 16. (b) Keccak uses column parity (θ), a fixed permutation ($\rho\pi$, one edge shown), and row neighbors (χ).

reported fraction is $(C_0 - C_{-i})/C_0$. The instrumented baseline and its ablated variant share inputs and configuration. These are diagnostic runs with intentionally invalid outputs, not accelerated implementations.

KEM has 144 permutations and 24 transforms for the selected known-answer input. DSA is restricted to key generation: stubbing hashing changes signing retries and can prevent verification from terminating. The deltas include dependent scheduling and memory effects; they motivate specialization of both primitives, with benefit tested on correct complete requests.

B. Register placement defines the communication

With lane l holding $a[l + 32k]$, $0 \leq k < 8$, the first three forward NTT layers exchange registers within one lane. The remaining layers exchange lane l with $lxord$: $d = 16, 8, 4, 2$ for seven-layer ML-KEM, and additionally $d = 1$ for eight-layer ML-DSA. Coalesced loads and stores surround a transform whose intermediate state remains in GPRs.

Mapping Keccak word $A[x, y]$ to lane $x + 5y$ exposes column parity (θ), rotate-and-gather ($\rho\pi$), and row logic (χ). Each 64-bit word occupies two RV32 GPRs; rotations may need both halves even when an instruction returns one. Figure 1 contrasts these neighborhoods with NTT pairs. Prior polynomial accelerators organize computation around memory rows or accelerator-owned storage [20]–[23]; here the architectural lane vector is the boundary. Fixed communication can replace generic shuffles while leaving sampling and request control on the programmable core.

C. From software references to matched baselines

Portable C provides byte-exact reference outputs and the historical diagnosis. PQRV’s optimized scalar RISC-V Kec-

TABLE II

RV32 COLLECTIVE ISA. ALL FORMS WRITE ONE rd PER ACTIVE LANE.
 $h \in \{L, H\}$ SELECTS A HALF; $Q \in \{K, D\}$ SELECTS A MODULUS.
 NTTBF ADDITIONALLY SELECTS CT/GS AND XOR DISTANCE 2^s .

Instruction	$rs1, rs2$	Result at lane l
KTHETA. h	A_L, A_H	$\theta(A)_h[l]$
KRHOPI. h	A_L, A_H	$\rho\pi(A)_h[l]$
KCHII. h	A_h, r	$\chi_{\iota_r}(A_h)[l]$
KROUND. $h r$	A_L, A_H	$\text{Round}_r(A)_h[l]$
NTTBF. Q	v, z	Paired CT/GS result
NTTMUL. Q	a, b	$\text{Mont}_Q(a_l b_l)$

cak [7] provides a separate scalar comparison, not the denominator of the main board results. Our primary baseline, A, uses the register layouts above with software shuffles and the same surrounding warp optimizations as the hardware arms. B adds Keccak Stage; C adds shared NTT; D adds both; E additionally reuses modular hardware outside transforms. A–E isolate hardware gains under one software mapping. Separate matched configurations evaluate Keccak granularity and multiplier-bank size (Section IV).

III. DESIGN AND IMPLEMENTATION

A. Collective ISA

On W32, each collective reads two source register vectors and writes one ordinary GPR per active lane. State persists in these GPRs; the instruction neither accesses memory nor retains a private cryptographic context. Keccak requires all 32 lanes active, with state in lanes 0–24 and zero results in lanes 25–31. NTT butterflies require a pair-closed mask: either both XOR partners participate or neither does.

Table II defines the instruction families. Keccak Stage exposes KTHETA, KRHOPI, and KCHII; KROUND fuses one round. RV32 L/H forms return one half of each 64-bit word. Theta, rho/pi, and Round read both old halves; chi/iota reads only its selected half and a warp-uniform round $r \in [0, 23]$. Round encodes r as an immediate. Software saves the first result until the second instruction has read the old pair (Fig. 3b).

For NTTBF, choose pair-low lane p with bit s clear and partner $h = p \text{ xor } 2^s$. The sources supply $a = v[p]$, $b = v[h]$, and $\zeta = z[p]$; the pair-high twiddle is unused. The forward CT form computes

$$t = \text{Mont}_Q(b\zeta), \quad (rd_p, rd_h) = (a + t, a - t). \quad (1)$$

Thus two source vectors express three butterfly inputs, while one destination vector carries both outputs. GS forms preserve each library’s inverse convention (Section III-C). Lane-local NTTMUL multiplies the two sources independently in every active lane. Stage and NTT use custom-0 (opcode $0 \times 0B$); function fields select operation, half, modulus, direction, and $s \in [0, 4]$. Round uses custom-2 ($0 \times 5B$), with its round $r \in [0, 23]$ in `func7`. All register dependencies are explicit.

B. Keccak Collective Datapath

Figure 2c implements fixed connections among the 25 state lanes. Theta first XOR-reduces each five-word column to $C[x]$, then injects $C[x - 1] \oplus \text{rol}(C[x + 1], 1)$ into that column. Rho/pi uses constant bit rotations and fixed word wiring. Chi combines three neighboring words in each row; iota XORs the selected round constant into word zero.

Stage has two pipeline registers. Column parity is captured at the first, and the following logic completes theta before the result register; rho/pi and chi/iota select their corresponding paths through the same interface. Round instead pipelines column parity, theta, rho/pi, and chi/iota across four stages. Both units accept one instruction per cycle without backpressure. Stage needs six collective issues per RV32 round, while Round needs two. Each Round half instruction recomputes the full internal round and selects its output half, keeping the two issues independent of hidden state. A separate memory-coupled engine reads and writes a 200-byte state as the Pointer comparison; it does not use the register-collective interface.

C. Shared K/D Modular Datapath

Figure 2b separates pair preparation, a shared signed 32×32 multiplier bank, modular reduction, and lane-result assembly. For CT, one Montgomery product supplies both lane results through add/subtract logic. Inverse GS preparation preserves the libraries’ different conventions:

$$\begin{aligned} \text{GS}_K(a, b, \zeta) &= (\text{Barrett}_K(a + b), \text{Mont}_K((b - a)\zeta)), \\ \text{GS}_D(a, b, \zeta) &= (a + b, \text{Mont}_D((a - b)\zeta)). \end{aligned} \quad (2)$$

K mode uses signed low-16-bit operands and arithmetic, then sign-extends its result; D mode uses signed 32-bit coefficients. The paths preserve native residue representations rather than forcing canonical residues at the boundary. Montgomery radices are 2^{16} for K and 2^{32} for D.

The same multiplier bank serves both moduli. K mode sign-extends its inputs and uses shift-add Montgomery constants; K-GS also sends the sum through a product with 20159 for Barrett reduction. D mode uses the full product with its own Montgomery constants. Mode selection controls operand preparation and reduction; capture, multipliers, beat scheduling, coefficient bypass, and result assembly remain shared. NTTMUL selects lane-local operands for this datapath, supporting inverse scaling, KEM base multiplication, and DSA pointwise products without a second modular unit. Twiddles arrive in $rs2$; there is no internal twiddle ROM or transform-layer state.

A bank of M physical multipliers serializes the logical products without changing instruction semantics or the binary. CT and D-GS require $B = 16/M$ beats. K-GS requires $B = 32/M$ because each pair needs both a Montgomery and a Barrett product; 32-lane NTTMUL also uses $32/M$ beats. Figure 3a shows M2 issuing 16 CT products in eight beats. Tagged products traverse the factor and reduction registers; indexed assembly collects the results and emits one vector after the last beat. Without stalls, the bank accepts an instruction every

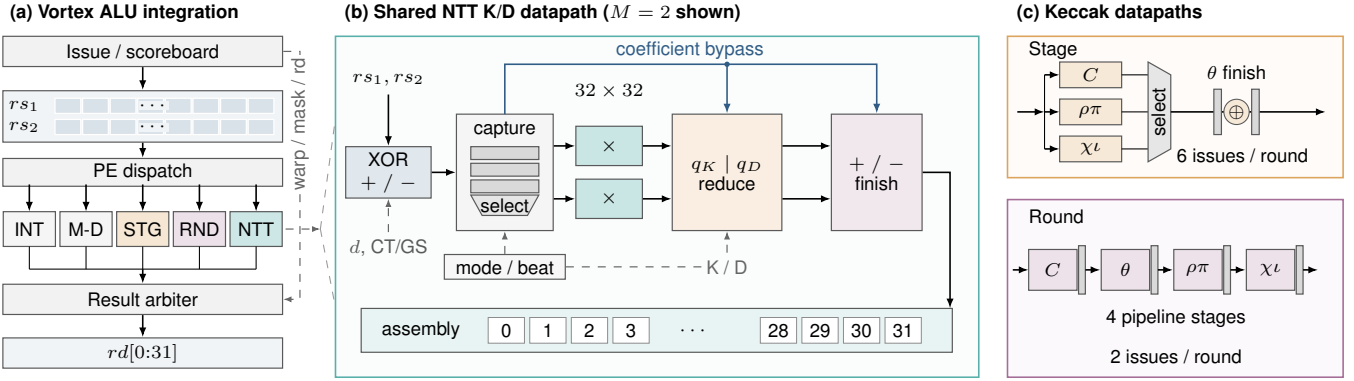
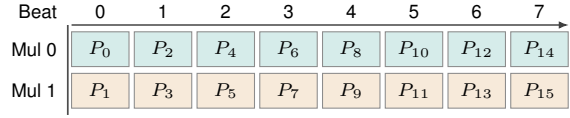


Fig. 2. Core integration and collective datapaths. (a) Dedicated PEs share operand dispatch and result arbitration with INT and multiply/divide. (b) Pair preparation includes GS pre-add/subtract. Both moduli share the multiplier bank, coefficient bypass, and result assembly. (c) Stage forms column parity C before its first register and completes θ after it; Round pipelines all four steps. Thin vertical bars denote pipeline registers.

(a) Two multipliers, sixteen pair products



(b) RV32 Round: preserve the old register pair

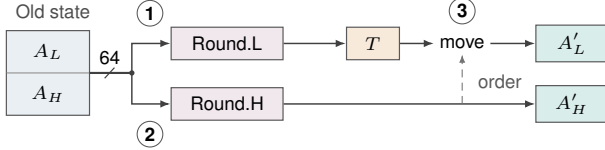


Fig. 3. Serialization and dependencies. (a) M2 assigns 16 pair products to two multipliers in eight CT or DSA-GS beats; pipeline latency is omitted. (b) Program order for an RV32 Round: both issues read the old halves, the low result remains in T , and the final move updates A_L .

B cycles; the measured ALU acceptance-to-commit latency is $B + 6$ cycles in this configuration. Physical capacity is a microarchitectural choice, evaluated separately in Section IV.

D. Execution Control and Core Integration

Figure 2a places Stage, Round, and NTT beside the integer and multiply/divide PEs. Decode names the ordinary source and destination registers for scoreboard tracking; ALU dispatch selects the PE, and a round-robin result arbiter feeds the existing commit path. No additional register-file port or private architectural register bank is required.

A serialized NTT instruction captures its operands and warp metadata on the first handshake. Saved operands feed subsequent beats; a new vector can be accepted once those beats have launched. Warp ID, destination, mask, and beat tags follow the products through the pipeline. Output backpressure freezes the serializer, pipeline, and partial-result assembly together, preserving identity across stalls. Each instruction writes back one vector exactly once. Other ready warps can issue to available PEs while its destination remains pending.

E. Software Integration

The native libraries retain their APIs and byte-exact outputs [24], [25]. Lane zero broadcasts hook arguments, enters the cooperative region, and resumes scalar control afterward. One-shot SHAKE keeps state in registers across absorb, permutation, and squeeze; incremental contexts are stored between calls. NTT loads eight coefficients per lane and completes the transform without intermediate memory. Lane-local butterflies use NTTMUL plus scalar arithmetic; XOR-paired butterflies use NTTBF.

The leader controls allocation and rejection loops, reclaiming temporaries after every signing attempt. Ininsics expose register operands to the compiler; cooperative kernels enforce the mask and old-half requirements of Section III-A.

IV. EVALUATION

A. Experimental Methodology

Table III specifies the V80 configuration: the NTT bank has two multipliers, the D-cache is write-through, and L2/L3 are disabled. All physical implementations use Vivado 2025.1 OPT3; the board clock reads back at 200 MHz before and after measurement. A request includes keypair/encapsulation/decapsulation for KEM or keypair/sign/verify for DSA. Device makespan excludes host transfers; we report the median of five runs after one warmup.

A–E retain the same warp sampling, KEM encoding/noise/linear arithmetic, and DSA SHAKE extraction, accumulation, and signing arithmetic. Zeroization uses the warp with one worker and the leader otherwise. Opcode audits verify no PQC instructions in A and no NTT instructions in A/B, including Montgomery conversion. Modes share the ordered inputs and board image. All 156 board cells (780 timed runs) pass byte-exact checks. Separate fusion and bank controls retain their archived write-back configurations, identified below.

The SimX timing model mirrors bank beats, pipeline latency, and stalls. RTL tests cover both moduli and directions, all XOR distances, masks, backpressure, and warp metadata.

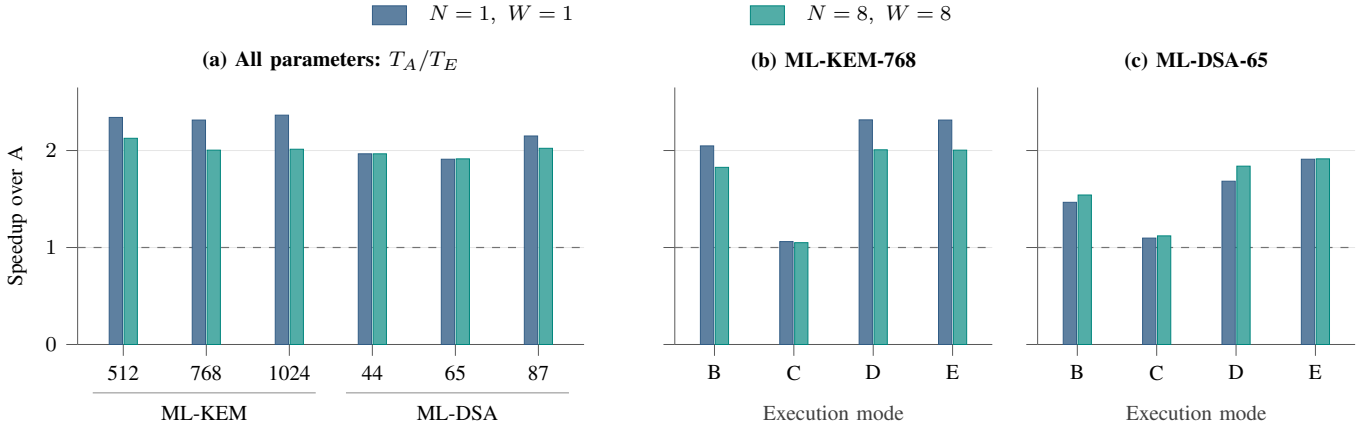


Fig. 4. 200-MHz V80 complete-request speedups over matched cooperative software (A). B enables Keccak Stage; C enables shared NTT; D enables both; E also uses modular products outside transforms. Blue/teal denote one/eight requests, with one worker per request. (a) A/E across all six sets; (b,c) B–E ablations for KEM-768 and DSA-65. Dashed lines mark software A at $1\times$. Every arm uses the same software mapping; ratios divide five-run median makespans.

TABLE III
PRIMARY EXPERIMENTAL CONFIGURATION.

Processor	RV32IM, one core, F/D disabled
Warp organization	8 resident warps, 32 lanes
On-core memory	16 KiB I\$, D\$, and local memory
Workloads	ML-KEM-512/768/1024; ML-DSA-44/65/87
Timing	XRT RTL integration; SimX controls; V80 silicon
Physical flow	Alveo V80, 200 MHz, Vivado 2025.1

XRT exercises the AFU; complete-library outputs are checked byte for byte against portable C. All 60 single/eight-request A–E cases across six sets retire identical SimX/XRT instruction counts with at most 4.422% cycle difference, below the fixed 5% bound. Their five board repetitions also match the XRT instruction counts.

B. Complete-Request Acceleration

Figure 4 uses a single complete-mapping cohort. At one request, E accelerates ML-KEM by $2.314\text{--}2.365\times$ and ML-DSA by $1.910\text{--}2.150\times$ over A. At eight requests, the ranges are $2.004\text{--}2.126\times$ and $1.914\text{--}2.023\times$, respectively. Accelerated single-request latency is 5.912–14.757 ms for KEM and 51.264–107.181 ms for DSA.

C. Contributions of the Hardware Paths

The ablation separates each hardware contribution. Across parameter sets and one/eight requests, Stage alone gives $1.827\text{--}2.124\times$ for KEM and $1.466\text{--}1.744\times$ for DSA (A/B). With Stage fixed, shared NTT adds $1.082\text{--}1.153\times$ for KEM and $1.118\text{--}1.194\times$ for DSA (B/D). NTT therefore adds request-level value after Keccak acceleration.

Reusing the modular unit outside transforms adds $1.040\text{--}1.146\times$ for DSA (D/E), with no new RTL. KEM’s corresponding range is only $0.998\text{--}1.024\times$; set 768 is within 0.2% of D. All headline results use E. Within a matched case, $(A/B)(B/D)(D/E) = A/E$; ratios from separate cohorts are not combined.

D. Granularity and Arithmetic Capacity

Figure 5(a) compares Round with equally unrolled Stage. At one and eight active warps, fusion improves dependent permutation throughput by $3.168\times$ and $3.291\times$, respectively. Separate SimX complete-KEM launch-cycle gains are only $1.028\times$ and $1.033\times$; the eight-request fusion control has no paired XRT run. A separate instrumented XRT control attributes only 4.964% of request time to Stage permutation; sponge I/O, sampling, and other work limit further gains. This profile does not describe the six-parameter board cohort. Fixed-mapping XRT controls also find Round $1.130\times$ faster than Pointer at one request, but Pointer $1.374\times$ faster at eight. Fusion and offload therefore require request-level evaluation at the intended concurrency.

Figures 5(b,c) distinguish test-program cycles from complete requests. The former uses historical RV32IMF W4T32 processor RTL, including loads, stores, and loops. The latter fixes Pointer Keccak in RV32IM W8T32 write-back XRT runs for KEM-768/DSA-65 at one/eight requests. Each timing experiment preserves its binary and inputs across banks.

M16 minimizes NTTBF test cycles. M8 and M4 increase them by 0.181% and 0.715%; M2 increases them by 2.963%, with at most 0.143% request overhead. The independent routed study in Fig. 5(d) finds that M2 removes 71.074% of NTT LUTs, 46.131% of FFs, and 87.500% of DSPs. M1 saves only five more core DSPs and increases NTT FFs, while test cycles rise by 25.043% over M16. M2 retains most resource savings with little measured request penalty.

All five NTTBF tests retire 59,534 instructions, pass 1,044 vectors, and show at most 0.959% SimX/RTL difference. Request sensitivity is much smaller than test-program sensitivity; identical instruction counts exclude a software change but do not isolate the contribution of latency hiding.

E. Concurrency and Batch Size

Figure 6 separates occupancy from batch size. On eight fixed inputs, increasing workers from one to eight raises

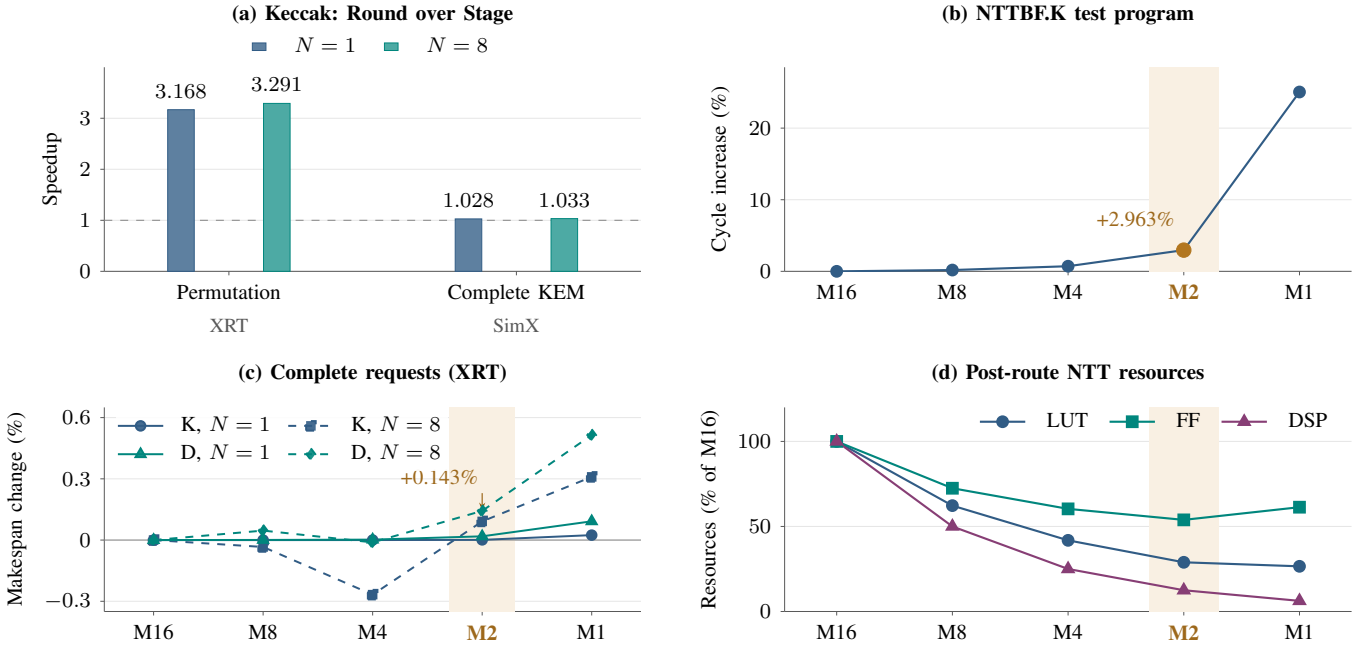


Fig. 5. Granularity and capacity in separate controlled studies. (a) Round/Stage: XRT permutation intervals and SimX KEM launch cycles, with $N = 1/8$ active warps. (b) Historical RV32IMF W4T32 NTTBF.K test-program cycles. (c) RV32IM W8T32 write-back XRT requests with Pointer Keccak: K=ML-KEM-768, D=ML-DSA-65. (d) Independent 200-MHz write-through NTT resources. Bank comparisons use M16 as reference; gold bands mark M2.

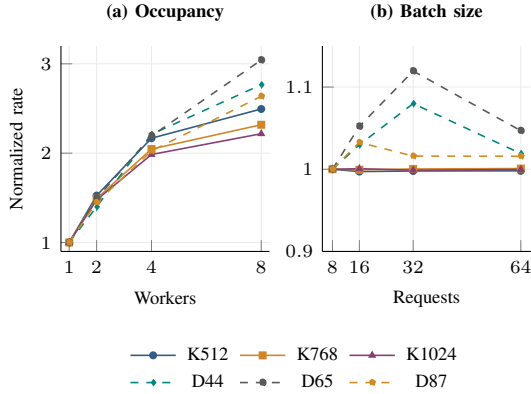


Fig. 6. 200-MHz board throughput for E. (a) Eight fixed inputs, normalized to one worker. (b) Eight workers, normalized to an eight-request batch. Curves cover all six parameter sets.

E throughput by $2.218\text{--}3.042\times$. With eight workers fixed, extending batches from eight to 64 changes throughput to only $0.998\text{--}1.047\times$ its eight-request value. Intermediate DSA batches peak at $1.119\times$, so throughput is not monotonic. Batches use synchronized waves. DSA signing retries vary by input; extending a batch also extends its input prefix, so its curve includes both amortization and workload variation.

F. Routed Cost and Design Scope

All cores in Table IV close without routing errors. M2 reduces whole-core LUTs/DSPs by $4.190\%/39.773\%$ over M16; Stage adds 1.289% LUTs to M2. At M16, Round uses

TABLE IV
INDEPENDENT RV32IM POST-ROUTE CORES AT 200 MHz. M16 VARIES
KECCAK GRANULARITY; M2 IS MEASURED BOTH ALONE AND WITH
STAGE.

Variant	LUT	FF	DSP	WNS (ns)
M16 base	233,759	136,451	176	+0.078
M16 + Stage	237,073	138,503	176	+0.116
M16 + Round	236,625	142,541	176	+0.052
M16 + Pointer	243,399	139,407	176	+0.080
M2 base	223,964	131,306	106	+0.048
M2 + Stage	226,852	133,368	106	+0.145

fewer LUTs but more FFs than Stage, so neither dominates in resource cost. The board AFU also includes optional engines: 243,073 LUTs, 181,881 FFs, 106 DSPs, and 0.001 ns setup slack. Board ratios compare software paths within this image.

Six-parameter results validate Stage+M2, without repeating the bank sweep or establishing joint optimality. Other transform intensity or concurrency may favor another design point. Energy and host-to-host latency are unmeasured.

V. CONCLUSION

Communication-matched collectives expose Keccak neighborhoods and NTT pairs through ordinary lane registers, with shared K/D modular arithmetic. All six parameter sets gain $1.910\text{--}2.365\times$ for complete single requests over cooperative software at 200 MHz. Independent granularity and bank studies connect primitive timing to request benefit and routed cost.

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